Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Generative AI for 3D Modeling

TGS Ref No: TGS-2023037544

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 8.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
8.0	15 August 2026	Rebuilt the course around Blender and an AI-assisted 3D modelling workflow; expanded concepts, activities, visuals and assessments.	Dr Alfred Ang

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Course Overview

A four-day, 32-hour WSQ programme in which learners use generative AI and Blender to create, validate, refine and release 3D assets aligned with technical-drawing knowledge and ability criteria.

Learning Outcomes

- LO1: Devise metrics for AI-assisted 3D design outputs and formulate solutions to modelling problems.
- LO2: Analyse Blender models for accuracy, topology quality and alignment with design requirements.
- LO3: Review and prepare 3D assets for organisational needs, applicable standards and production delivery.

Daily Schedule

Day	Time	Duration	Topic / Activity	Method	Slides
1	09:00–10:30	1h 30m	Course administration, outcomes and AI-assisted 3D workflow	Facilitated briefing and visual discussion	Slides 2–19
1	10:30–12:30	2h	Topic 1: briefs, prompts, references and Blender scene foundations	Concept demonstration and guided analysis	Slides 20–84
1	12:30–13:30	1h	Lunch	Break	—
1	13:30–15:30	2h	Activities 1–2: model brief and AI-assisted Blender script	Hands-on practice	Slides 68–75
1	15:30–18:00	2h 30m	Topic 1: mesh anatomy, modifiers, bpy and safe iteration	Demonstration, practice and review	Slides 20–84
2	09:00–10:30	1h 30m	Topic 1: design variation, provenance and responsible use	Frameworks and worked examples	Slides 20–84
2	10:30–12:30	2h	Activities 3–4: script repair and variant selection	Hands-on practice	Slides 76–83
2	12:30–13:30	1h	Lunch	Break	—
2	13:30–15:30	2h	Topic 2: validation, silhouette and topology	Demonstration and guided inspection	Slides 85–150
2	15:30–18:00	2h 30m	Activities 5–6: baseline inspection and local repair	Hands-on practice	Slides 134–141
3	09:00–10:30	1h 30m	Topic 2: retopology, budgets and UV strategy	Frameworks and worked examples	Slides 85–150
3	10:30–12:30	2h	Activity 7: retopology and optimisation	Hands-on practice	Slides 142–145
3	12:30–13:30	1h	Lunch	Break	—
3	13:30–15:30	2h	Topic 2: PBR materials, AI textures, lighting and change control	Demonstration and review	Slides 85–150
3	15:30–18:00	2h 30m	Activity 8 and Topic 3 introduction	Hands-on practice and facilitated discussion	Slides 146–149; 151–152
4	09:00–10:30	1h 30m	Topic 3: standards, target-use workflows, rendering and export	Frameworks and demonstrations	Slides 151–215
4	10:30–12:30	2h	Activities 9–10: evidence renders and real-time export	Hands-on practice	Slides 199–206
4	12:30–13:30	1h	Lunch	Break	—
4	13:30–15:00	1h 30m	Activities 11–12: print preparation and capstone release review	Hands-on practice and course wrap-up	Slides 207–214
4	15:00–16:00	1h	Written Assessment (SAQ)	Open-book assessment	—
4	16:00–18:00	2h	Practical Performance (PP)	Open-book practical assessment	—

Topic-by-Topic Breakdown

Topic 1: Generative AI for 3D Design Concepts and Modeling · Slides 20–84

Briefs · prompts · Blender foundations · script-assisted construction · design variants

Alignment: K1 · K2 · K3 · K6 · A1 · A2

Activity 1: Translate a Design Requirement into an AI-Ready 3D Brief · Slides 68–71

Create measurable geometry, appearance and delivery criteria before using AI.

Activity 2: Generate a Blender Gold-Sphere Scene with AI-Assisted Python · Slides 72–75

Use the video workflow to request, inspect, run and verify a small Blender Python script.

Activity 3: Iterate and Repair an AI-Generated Blender Script · Slides 76–79

Diagnose a controlled script failure and use evidence-driven AI feedback to repair it safely.

Activity 4: Develop and Select Three AI-Assisted Product Variants · Slides 80–83

Generate meaningful design alternatives and select one using weighted acceptance criteria.

Topic 2: AI-Assisted 3D Model Analysis, Validation and Refinement · Slides 85–150

Inspection · topology · normals · retopology · UVs · materials · controlled repair

Alignment: K1 · K4 · K5 · A2 · A3 · A4

Activity 5: Inspect an AI-Generated Mesh and Build a Validation Baseline · Slides 134–137

Measure geometry, transforms, topology and evidence before making repairs.

Activity 6: Repair Non-Manifold Geometry, Normals and Duplicate Data · Slides 138–141

Apply local, reversible repairs and prove that the original failing checks now pass.

Activity 7: Retopologize and Optimize an AI Mesh for Its Target Use · Slides 142–145

Select a retopology approach, meet a polygon budget and preserve critical form.

Activity 8: UV-Unwrap and Apply AI-Assisted PBR Materials · Slides 146–149

Create usable UVs and physically plausible materials from AI-assisted references.

Topic 3: 3D Model Optimization, Standards and Production Workflows · Slides 151–215

Standards · rendering · export · interoperability · documentation · capstone delivery

Alignment: K3 · K4 · K5 · K6 · A3 · A5 · A6

Activity 9: Create a Neutral Studio Render and Visual QA Set · Slides 199–202

Use consistent cameras, lighting and colour settings to create comparable validation evidence.

Activity 10: Prepare and Verify a Game/Visualization Export · Slides 203–206

Package a model for real-time delivery and verify it through round-trip import.

Activity 11: Prepare and Verify a 3D-Printable STL · Slides 207–210

Convert a model into a watertight, correctly scaled print candidate with manufacturability evidence.

Activity 12: Capstone: Deliver an AI-Assisted 3D Asset Review Package • Slides 211–214

Integrate brief, AI assistance, Blender refinement, validation, export and stakeholder recommendation.

Resources Required

- Blender installed on Windows or macOS laptop
- AI assistant able to draft text and Blender Python
- Course activity folders and sample Blender assets
- Presentation software for activity evidence
- Internet access for course references and LMS submission

Assessment

Written Assessment (SAQ): 6 open-ended knowledge questions, 1 hour, open book.

Practical Performance (PP): 3 integrated Blender tasks, 2 hours, open book.

3 hours on Day 4, from 3:00 PM to 6:00 PM.

Assessment is individual and open book. Learners complete assessment digital attendance, submit on the LMS and sign the Assessment Summary Record.